Industrial IoT Equipment Health Monitoring System

Abstract

Industrial equipment failures can lead to production downtime, increased maintenance costs, and reduced operational efficiency. This project presents an Industrial IoT Equipment Health Monitoring System using ESP32, DHT22 sensors, and vibration monitoring. The system continuously monitors machine health parameters such as temperature, humidity, and vibration levels. When abnormal conditions are detected, an alert is generated through LED indicators. The complete system is simulated using Wokwi and demonstrates the application of IoT in industrial monitoring.

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1. Introduction

Industrial machines operate continuously under varying environmental conditions. Excessive temperature and vibration are common indicators of potential machine failures. Traditional monitoring methods require manual inspection and may not provide real-time information.

The proposed Industrial IoT Monitoring System automates machine health monitoring using sensors connected to an ESP32 microcontroller. The collected data is processed in real time to identify abnormal operating conditions.

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2. Problem Statement

Manual monitoring of industrial equipment is time-consuming and prone to human error. Unexpected machine failures can result in significant financial losses and downtime.

Therefore, a low-cost IoT-based monitoring system is required to continuously monitor machine health and provide alerts when operating conditions exceed safe limits.

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3. Objectives

- Monitor temperature and humidity using DHT22.
- Simulate vibration monitoring using a potentiometer.
- Detect abnormal machine conditions.
- Provide visual alerts using LEDs.
- Demonstrate Industrial IoT concepts using ESP32.
- Simulate the complete system using Wokwi.

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4. Components Used

Hardware Components

Component	Quantity
ESP32 Dev Module	1
DHT22 Sensor	1
Potentiometer	1
Green LED	1
Red LED	1
Resistors	2

Software Tools

- Wokwi Simulator
- Arduino IDE
- GitHub
- ESP32 Libraries
- DHT Sensor Library

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5. System Architecture

Machine Parameters

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DHT22 + Vibration Sensor

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ESP32 Controller

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Fault Detection Logic

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Status LEDs

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User Monitoring

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6. Working Principle

The ESP32 continuously reads data from the DHT22 sensor and vibration sensor.

The following conditions are monitored:

- Temperature Threshold = 50°C
- Vibration Threshold = 2500 ADC Value

If either threshold is exceeded:

- Red LED turns ON
- Green LED turns OFF
- Alert message is displayed on Serial Monitor

Otherwise:

- Green LED remains ON
- System status is displayed as NORMAL

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7. Circuit Connections

DHT22

- VCC → 3.3V
- GND → GND
- DATA → GPIO15

Potentiometer

- VCC → 3.3V
- GND → GND
- SIG → GPIO34

Green LED

- Anode → GPIO18
- Cathode → GND

Red LED

- Anode → GPIO19
- Cathode → GND

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8. Results

Normal Condition

Temperature below threshold and vibration within safe limits.

Output:

Machine Status: NORMAL

Green LED ON

Fault Condition

Temperature greater than 50°C or vibration greater than threshold.

Output:

ALERT: MACHINE HEALTH ISSUE DETECTED

Red LED ON

The system successfully identifies abnormal operating conditions and provides immediate visual feedback.

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9. Advantages

- Low-cost implementation
- Real-time monitoring
- Easy to expand
- Industrial relevance
- Demonstrates IoT concepts

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10. Applications

- Manufacturing Industries
- Predictive Maintenance Systems
- Smart Factories

- Industrial Automation
- Equipment Health Monitoring

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11. Future Enhancements

- MQTT Integration
- Cloud Connectivity
- Node-RED Dashboard
- Grafana Visualization
- SMS and Email Alerts
- Predictive Maintenance using Machine Learning

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12. Conclusion

The Industrial IoT Equipment Health Monitoring System successfully demonstrates real-time monitoring of machine parameters using ESP32. The project provides a foundation for developing advanced Industrial IoT solutions and predictive maintenance systems. The use of Wokwi simulation enables rapid development and testing without requiring physical hardware.

References

1. ESP32 Technical Documentation
2. DHT22 Sensor Datasheet
3. Arduino Official Documentation
4. Wokwi Simulation Platform Documentation
5. Industrial IoT Research Articles